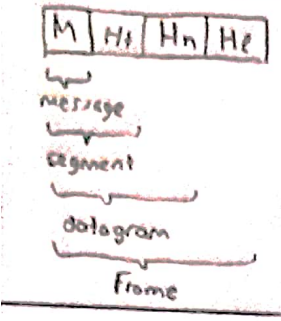
The packet information at the **application** layer is **message**.
The packet information at the **transport** layer is **segment**.
The packet information at the **network** layer is **datagram**.
The packet information at the **link** layer is **frames**.

when sender sends 100 bits/sec and receiver receives 50 bits/sec thats a flow control issue. It is a local issue happens between two end systems.
However, congestion control is global issue.
Congestion control makes sure network able to carry offered traffic. It includes all hosts etc.



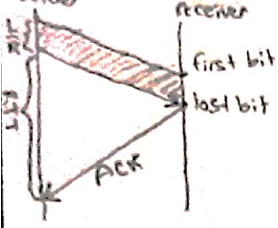
Application → SMTP, HTTP, FTP, POP3, SNMP
Transport → TCP, UDP
Network → IPv5, IPv6, ICMP, IPSEC, ARP, MPLS
Link → Fiber cable, RAPA etc., ISDN

Demultiplexing at receiver use header info to deliver received segments to correct socket
Multiplexing at sender add transport layer header. Send data from one of sockets

Neither TCP nor UDP provides delay guarantees or bandwidth guarantees.

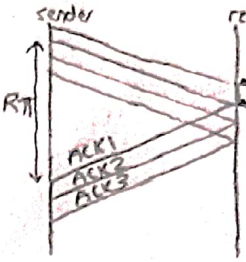
NAK receiver explicitly tells sender that packet had errors.
ACK or NAK may also be corrupted.

RDT means Reliable data transfer. Stop and wait sender send one packet, then waits for receiver response.
In case ACK or NAK is corrupted, we use sequence number for each packet. Receiver discards duplicate packet. (0,1) suffice.
We would like to be NAK free, so isACK(rcvpkt, 0) same as isACK(rcvpkt) and isACK(rcvpkt,1) same as isNAK(rcvpkt)
But packets may also be lost and we would never receive NAK, ACK etc. in this case sender waits some time...
We stop and start the timer for particular transitions.



$$U_{sender} = \frac{\frac{L}{R}}{RTT + \frac{L}{R}}$$

If we use pipelining:



$$U_{sender} = \frac{3L/R}{RTT + \frac{L}{R}}$$ = utility tripled.

Go-Back-N uses a window to send packets in order. 0 1 2 3 4 5 6 7 8.

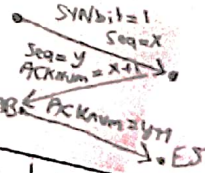
Selective Repeat sender maintains a timer for each unACKed packet. It individually acknowledges all the packets.

- Selective Repeat does not discard the packets which are previously ACKed in window.
- They are buffered on receiver side. Go-Back-N discards them.
- If earlier packets not yet arrived, windows will not advance and later packets are buffered.

TCP is flow controlled. Sender will not overwhelm receiver.
TCP: ACK and Seq numbers cross incremented by one.
IF time outs are short, retransmissions will increase.
RTT (SampleRTT) measured average time from segment transmissions until ACK receipt.

EstimatedRTT = $(1-\alpha)$ · EstimatedRTT + α SampleRTT So this is how TCP recomputes the estimated RTT each time a new SampleRTT is taken. It is called EWMA (Exponential weighted moving average). Typical value $\alpha = 0.125$
So timeout value is taken to be TimeoutInterval = EstimatedRTT + 4× DevRTT where DevRTT is standard deviation.
DevRTT = $(1-\beta)$ DevRTT + β |SampleRTT - EstimatedRTT|

In TCP retransmit, we assume it is likely that unACKed segment lost, so don't wait for timeout. Just resend unACKed segment.
- So how does TCP provide flow control? [Receive Window] is used this point. (rwnd field in TCP header)
many operating system auto adjusts this buffersize.
- How TCP handshakes? "Hey there!" and "I am here!" messages not enough. Instead:



So we need extra ACK.

Due to retransmissions, the maximum end to end throughput can actually be significantly less than the actual transmission capacity of a congested router.
λ_{in} original data transmission rate, λ'_{in} data plus retransmitted data transmission rate.
Retransmission is every queues problem. It is global issue. It reduces throughput everywhere.
In TCP, sender informs receiver about loss and delays. Then necessary actions are taken.

QUIC:
Quick UDP connections.

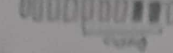
AIMD is used for TCP congestion control.

Additive Increase: increase sending rate by one segment

Multiplicative Decrease: cut sending rate in half at each loss event.

TCP cuts in half when loss detected by triple duplicate ACKs.

cwnd: congestion window

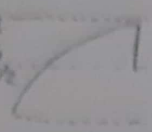


$\text{TCP Rate} \approx \frac{\text{cwnd}}{\text{RTT}}$ bytes/sec cwnd is dynamically adjusted.

slow start means double cwnd for each successful send.

TCP CUBIC is looks similar to AIMD. R

TCP CUBIC is used in Linux.



Delay-based TCP congestion control aveve is "just full enough but no fuller". Uncongested throughput $\frac{\text{cwnd}}{RTT_{min}}$

Explicit Congestion Notification (ECN) two bits in IP header (ToS field) marked by network router to indicate congestion.

TCP fairness goal: If k TCP sessions share same bottleneck link of bandwidth R, each should have R/K rate.

Data Plane where to forward?

Control Plane which routes etc from source to destination.

Traditional networking means the routing algorithms are implemented in routers.

Software defined networking (SDN) implemented in remote servers.

Best effort service model has no guarantees for: Loss, Bandwidth, Order, Timing... (simplicity)

Destination-based forwarding forward based only on destination IP address.

Generalized forwarding forward based on any set of header field values.

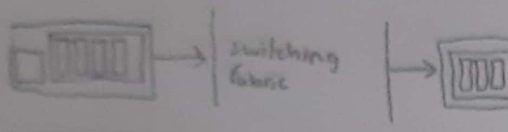
IP based forwarding uses bit ranges to determine the output port.

Longest Prefix Match selects output for IP, where prefix bit matches are the longest one. It uses TCAM ternary content adressing.

If switch fabric slower than input ports combined it will cause **input port queuing**.

Head of the Line (HOL) blocking: Queued datagram at front of queue prevents others in queue from moving forward.

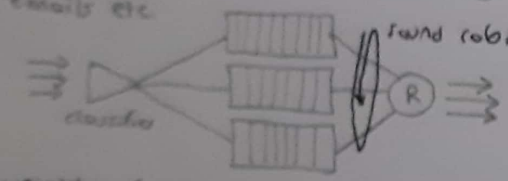
Output port Queueing when datagrams are arriving from switching fabric faster than the link transmission rate



There are scheduling and drop policies at this point.

$\frac{RTT \cdot C}{\sqrt{N}}$ is used for determining buffering size. Dropping can be done by using tail drop or priority drop.

Queueing can be done by using priority queues or FIFO queues. These priorities are defined by ISPs. Voices by be faster than emails etc.



weighted fair queuing each class i has weight w_i and gets amount of service by formula $\frac{w_i}{\sum w_j}$

For 32 bit IP we have following:

223.1.1.1 = 11011111 00000001 00000001 00000001

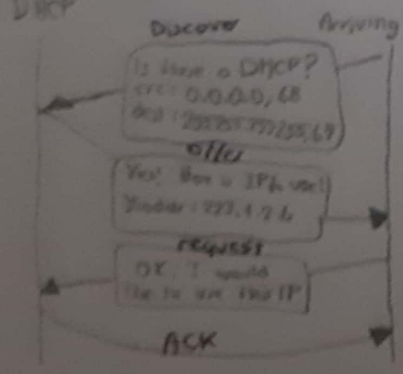
Interfaces may be WiFi or Ethernet cables.

Subnets are device interfaces that can physically reach each other without passing through an intervening router. Computers connected by LAN forms a subnet.

IP addresses have subnet part and host part. Left 24 bits form subnet part (high order bits) and remaining low order bits define the host part. 24 bits would mean first three terms being equal.

CIDR (Classless Inter Domain Routing) uses format: a.b.c.d/x where x is number of bits in subnet portion address.

How does host get an IP address? **DHCP Dynamic Host Configuration Protocol** is used. Host dynamically obtains IP address from network server when it "joins" network. DHCP located in a router can serve all subnets connected by this router.



Here, Discover and offer are optional. Host may also need IP of DNS server, address of first-hop router for client.

We discussed host. What about Network? How it can get an IP?

In IP address, first 20 blocks are for ISPs. ISPs can we these bits to determine the organizations. When a organization moves one ISP to other ISP, the new ISP can use longest prefix matching for the organization. How does an ISP get block of address? Answer: ICANN

NAT Network Address Translation all devices in local network share just one IPv4 address as far as outside world concerned. When we leave an local network through an NAT router, we will have same IP address. (32 bit) That way hosts can easily change IPs. To provide all these NAT translation table mapping WAN side addr → LAN side addr

S: 80.0.0.1, 7345 → outside → S: 138.76.29.7, 5001
D: 128.119.40.186, 80 → D: 128.119.40.186, 80

(destination same)

S: 128.119.40.186, 80 → inside → S: 128.119.40.186, 80
D: 138.76.29.7, 5001 → D: 10.0.0.1, 7345

(source same)

IPv6 provides larger address space, speed processing & forwarding.

ver	pri	flow label	
payload len	next header	hop limit	
source address (128 bits)			
destination address (128 bits)			
payload (data)			

IPv6 datagram format

whats missing in IPv6?
- No checksum
- no fragmentation/reassembly
- no options

So how will network operate IPv4 and IPv6 routers? **Tunnelling.** IPv6 datagram carried payload in IPv4 datagram among IPv4 routers.

Generalized forwarding uses **OpenFlow** where there is match, action, stats fields. Matching can be done in link, transport, network layers. A fire wall can permit packets where TCP d-port is 22. NATs rewrite source and destination addresses. So they are also using generalized forwarding where the action is changing the header. Router can also use "match plus action". It may forward packets by matching same address fields.

Middle Boxes consists of NATs, Firewalls, IPS, Caches, CDNs etc they are everywhere. Middle Boxes are located in Network layer.

Congestion control, retriability can be implemented in both network edges or network itself.

Control plane mainly stresses about network algorithms etc. Remember: Data plane stresses about forwarding. Control plane stresses about routing.

SDN remotely controls forwarding table of routers. A path is defined as sequence of routers
- **Global knowledge** algorithms ie **link state** algorithms have cost of all links etc (Centralized)
- **Decentralized** algorithms ie **distance vector** algorithms have costs for local links.

If routes changes slowly over time static, otherwise dynamic.

When link costs depend on traffic volume **route oscillations** possible.

Bellman-Ford is decentralized algorithm performing:

$$D_x(y) = \min_v \{C_{x,v} + D_v(y)\}$$

In the end, Bellman-Ford is able to find the least cost path for each source and destination. If changes in a link cost happens all routers computes new distance vectors and pass it to the neighbouring routers.

Count to infinity problem occures when link cost of some link increases for the distance vector algorithm.

Black holing when a route malfunctions and sends wrong cost informations whole network may be effected.

In reality we cant hold information about all routers. We parse routers into regions known as **autonomous systems (AS)** aka. **domains.** Internet is network of networks.

We have two routings for clustered routers: Intra-AS and Inter-AS.

Intra-AS routing among routers which are in same AS.

Inter-AS routing among ASes.

Gateway routers are connecting ASes. They are edge of Intra-Ases.

- Intra-AS Protocols

RIP (Routing Information Protocol)

OSPF (Open Shortest Path First)

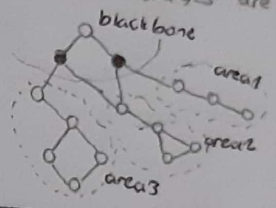
EIGRP (Enhanced Interior Gateway Routing Protocol)

RIP Deprecated Distance Vector routing where 30 seconds updates.

OSPF Link state routing. Also called ISIS (Dijkstra)

EIGRP Another Distance Vector routing protocol.

All OSPF messages are authenticated.

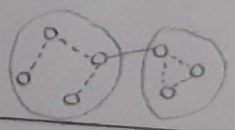


Link state ads only advertised within the area.
- are backbone routers.
- routers within area are called 'internal' (local routers)

BGP Border Gateway Protocol only protocol that is for InterAS. It is defacto and glue of internet.

eBGP obtain reachability info from neighboring ASes.

iBGP propagate reachability info to all AS-internal routers.



/ eBGP and / iBGP.

BGP, ASes share information. (TCP179). AS3 promises to AS2 it will forward datagrams towards X. BGP messages exchanged between peers over TCP connection. These messages include OPEN, UPDATE, KEEPALIVE, NOTIFICATION

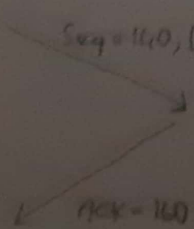
Sometimes ISPs will not desire to serve the traffic coming from other ISPs. So they do not advertise paths to these ISP routers.

So how forward tables are populated among ASes? **Hot potato routing** ignores inter costs and only considers intra costs. It chooses local gateway that has least intra domain cost. We would like to carry out the packet out of AS as soon as possible.

ICMP (Internet Control Message Protocol) is used by hosts and routers to communicate in network level.

ACK 300 demek toplamda 300 e kadar olan byteları aldım demektir. SEQ 300 yani toplamda 300 byte gönderdim ai biraz daha gönderiyorum demektir. Bu bu ACK ve SEQ degerleri cift yonlu hesaplanmasi gerekebilir.

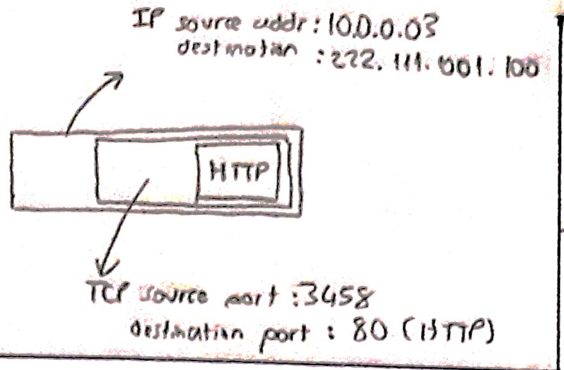
TCP timer her bir paket için çalıştırmaz.



eger bir sonraki paket gelirse discard edip eski paketin ACK'ini gondeririz.

Poisoned reverse penalties some links infinity.

IPV6 te options field yok çünkü header size constant. Constant olduğu için header size fieldde yok.

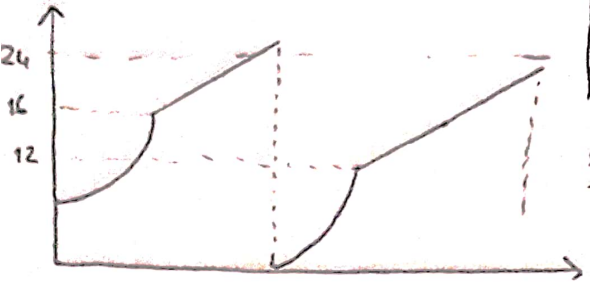


ICMP error reporting ve echo için kullanılıyor.
Başlangıçta TTL=1 set ediyorsun. İlk router için bunu kullanıyorsun.
TTL=2 ikinci router, TTL=3 üçüncü router destination router geldiğinde TTL bilinmiş olacaktır.

IPv4 header 20 böyle kabul ediyoruz. Eğer bir linkte 2000 MTU varsa bu linkteki datagram 4 e bölünmüşse:

1980 +20
1980 +20 şeklinde min
1980 +20
1 +20

1980 +20
1980 +20 şeklinde max
1980 +20 parçalama olur.
1980 +20

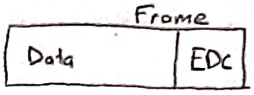


ssthresh is halved. We apply slow start until we hit ssthreshold. Then we start increasing linearly.

SNMP protocol is an network management protocol and it shouldnt be wasted by congestion control checks etc so it uses UDP.

In a single link, Media Access Control (MAC) is used to navigate packets/frames from destination and source addresses. Link layer implemented on-chip or in network interface card (NIC).

EDC Error Detection and correction bits



EDC can be a single parity bit, setting to 1 so there is an even number of 1s. Receiver checks if even number of set bits including EDC.

Two-D parity is also possible. Both rows and columns should have even number of set bits.

With the help of 2D parity check, we can correct single bit errors without retransmission!

EDC can also be the checksum. But CRC is most popular.

Cyclic Redundancy Check (CRC) data bits D and pattern G.
G is r+1 bit. We concatenate <D, R> such that:

$D \cdot 2^r$ (XOR) R.



Receiver knows G divides <DR>. If non zero rem then error!

$D2^r$ (XOR) $R = nG$ $\quad D2^r = nG$ (XOR) R

so: R = remainder $\left[\frac{D2^r}{G}\right]$. We use binary division.

Links can be point to point or broadcast etc.
If we receive two signal at the sametime collisions occur.
multiple access protocol determines when node can transmit via a link.
In multiple access protocol nodes takes turn to transmit
channel partitioning is also possible. (frequencies, time slots etc.)

TDMA we discussed it in chapter 1.
Time division multiple access.
FDMA frequency division multiple access.
Random access protocols let collision happen. Then fix it later.
Examples includes ALOHA, CSMA/CD, CSMA/CA.

ALOHA If no collusion success otherwise node retransmits frame each subsequent slot with probability p until success.
Purely taken as %37.

CSMA listen before transmit more polite than ALOHA.
if channel idle send, otherwise defer transmission.

CSMA/CD: CSMA with collision detection.
So collisions still occur. Because of propagation delay.
Ethernet uses CSMA/CD. Bluetooth uses taking turns.

DOCSIS Data over Cable Service Interface Specification.

MAC is 48 bits e.g. 71-65-F7-2B-08-53.
MAC does not have longest prefix matching.
Instead we use exact or complete matching.

ARP How to determine interface's MAC address, knowing its IP address? ARP table each IP node/MAC mappings.
<IP, MAC, TTL>

Suppose A doesnt know MAC of B.
A first sends broadcast message FF-FF-FF-FF-FF-FF.
All nodes on LAN receive ARP query. B responds, giving its MAC address. Then A updates its ARP table.
When A and B is not in same subnet this does not work.
Router does not forward ARP requests. So what we do?
At this point ARP request converted to datagram.
Most probably first hop router cached MAC of B.
Rest is easy.